

Nanosecond-level optical switching for flexible and resilient fronthaul networks

VAIGAI NAYAKI YOKAR,[†] ALI MEHRPOOYA,[†] SEN SHEN,^{id} YIRAN TENG,^{id} WANXIN ZHAO, SHUANGYI YAN,^{*} ^{id} AND DIMITRA SIMEONIDOU

Smart Internet Lab, University of Bristol, Bristol, UK

[†]These authors contributed equally to this work.

^{*}Shuangyi.yan@bristol.ac.uk

Received 12 September 2025; revised 14 December 2025; accepted 16 December 2025; published 28 January 2026

This paper presents a novel, to the best of our knowledge, ultra-low latency fronthaul networking approach based on nanosecond (ns)-precision optical switching with advanced synchronization. We propose and demonstrate a comprehensive control architecture that integrates precision time protocol (PTP)-based time alignment with semiconductor optical amplifier (SOA)-based optical switches, enabling coordinated, jitter-minimized switching with accuracy as fine as 100 ns. Two primary synchronization methods are explored: a distributed hardware approach employing dedicated timing devices for simultaneous switch configuration, and “SyncNet,” a software-defined networking (SDN) controller utilizing time information inspired by reverse PTP. Experimental validation on a commercial Open RAN 5G system shows that our fronthaul switching framework achieves exceptional reliability, evidenced by only 0.00024% packet loss during link recovery while maintaining steady throughput (uplink: 1.03 Mbps and downlink: 1.01 Mbps) over repeated switching events. The proposed architecture supports dynamic reconfiguration for rapid link recovery and energy-efficient, traffic-adaptive resource allocation. Our results establish a robust foundation for next-generation fronthaul networks, delivering deterministic performance and flexibility aligned with 6G requirements. This work directly addresses the escalating needs for ultra-low latency, high reliability, and energy efficiency in emerging mobile and AI-driven applications. © 2026 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

<https://doi.org/10.1364/JOCN.578768>

1. INTRODUCTION

The emergence of new mobile applications, particularly those powered by large language models (LLMs) and AI-generated content [1,2], is accelerating the evolution of 6G networks across both technological and commercial dimensions. These computationally intensive applications, such as real-time language translation, intelligent virtual assistants, and immersive XR experiences, demand a flexible and dynamically adaptable 6G infrastructure capable of meeting stringent performance requirements in terms of latency, bandwidth, and reliability [3]. According to the ITU-T’s IMT-2030 framework report published in 2023, 6G technology is expected to deliver peak data rates ≥ 1 Tbps, at least 50 times faster than 5G capabilities, with a delivery latency ≤ 0.1 ms, representing a tenfold improvement over 5G under all operation conditions [4,5]. Comprehensive industry forecast projects exponential growth in connected devices, with estimates reaching 500 billion IoT devices by 2030, generating unprecedented volumes of data traffic and driving demand for ultra-low latency, high-bandwidth applications [6].

In parallel with these application-driven demands, the Open radio access unit (RAN) ecosystem is gaining momentum, promoting disaggregated network functions and flexible deployment of network components across centralized units (CUs) and distributed units (DUs) [7]. This disaggregation approach enhances openness, interoperability, and vendor diversity across the RAN, potentially reducing deployment cost and fostering innovations. However, it also introduces complex architectural challenges, particularly in the fronthaul segment connecting DUs and radio units (RUs). While Open RAN successfully implements Ethernet-based packet transport for CU–DU midhaul connections, the fronthaul presents more stringent requirements [8]. Fronthaul, the transport segment between DUs and RUs, carries time-sensitive user-plane and control-plane information in 5G/6G radio access networks. This segment is distinct from backhaul and midhaul and is characterized by much tighter latency, jitter, and synchronization constraints—typically requiring nanosecond (ns)-level timing accuracy. These stringent requirements arise because fronthaul directly impacts radio signal processing and physical layer coordination. Even minor timing deviations can degrade

link performance or disrupt service continuity. As networks evolve toward dynamic and energy-aware architectures, precise timing and synchronization become critical enablers for fast reconfiguration and resilience in fronthaul transport. Currently, most deployments rely on static point-to-point connections between DUs and RUs due to strict latency and synchronization requirements under different functional split options. This static configuration becomes problematic, as both DU and RU are among the most power-hungry components in the RAN infrastructure, with energy consumption highly correlated with actual traffic load [9,10]. As 6G services evolve to increasingly dynamic and bursty traffic patterns, a more flexible fronthaul architecture becomes essential not only to accommodate traffic variability but also to enable energy-efficient resource allocation and rapid reconfiguration in response to changing network conditions for better network resilience [11]. Current exploration in the telecommunication industry has been focusing on several switching technologies to address the growing demands of the fronthaul section in terms of flexibility and reconfiguration [12]. Conventional Ethernet switching, while widely deployed for its cost-effectiveness and simplicity, introduces unpredictable latency variations and lacks the deterministic performance guarantees necessary for coordinated network operations in advanced fronthaul scenarios [6]. Flexible Ethernet (FlexE) offers dynamic bandwidth allocation through layer separation between MAC and PHY layers, enabling more efficient resource utilization compared to standard Ethernet [13]. However, FlexE implementation requires specialized equipment that significantly increases deployment costs and operational complexity [14]. Based on a similar concept, time-sensitive networking (TSN) enhances standard Ethernet with precise timing synchronization and deterministic packet delivery, achieving microsecond-level (μs) precision through features such as time-aware scheduling and frame preemption. Despite these capabilities, TSN deployment faces substantial challenges, including the need for specialized hardware throughout the network and potential congestion issues affecting high-priority traffic during peak loads [15]. This electrical solution also requires optical-electrical-optical (OEO) conversion for any switching operations. In summary, these electrical Ethernet-based switching solutions primarily focus on providing flexible, packet-level connectivity, and bandwidth management, but they rely on buffering, queuing, and OEO conversions and are therefore not well suited to forming a deterministic, multi-hop fronthaul network with real-time, low-latency switching.

Recently, there has been increased interest in bringing optical switching to the fronthaul section, considering the low-latency and fast switching capability [16–20]. Optical switching eliminates the power-consuming OEO conversions at switch nodes, improving energy and cost efficiency. Technological advancements in fast optical switches with ns-switching times, such as semiconductor optical amplifier (SOA)-based optical switches, have been exploited to realize data centre networks (DCNs) with high network throughput, flexible connectivity, and on-demand resource utilization [21–23]. Despite the promising potential of fast optical switching technologies, practical implementation of ns-level optical switching still faces several critical challenges [24], including

control, management, and scalability. The fronthaul switch solution requires innovations, including both the hardware switching and the associated controlling and management.

In this paper, we present, for the first time, the implementation and demonstration of a fronthaul network featuring enhanced flexibility to support DU migration and improved resilience for rapid recovery of fronthaul links (including both uplink and downlink). The RAN system is implemented based on the precise time protocol (PTP), synchronized fast optical switches. Our solution encompasses full control of 1×4 optical switches with a 10 ns switching time, PTP-based time synchronization, achieving tracking errors around 30 ns over several minutes, and a fully manageable, time-aligned optical switching infrastructure, enabled by “SyncNet,” a time-aware software-defined networking (SDN) controller. Leveraging these technologies, we deploy and validate the proposed fronthaul network using commercially available Open RAN 5G equipment. Two scenarios are demonstrated: (1) DU migration triggered by traffic dynamics through on-demand fronthaul reconfiguration, and (2) fast fronthaul link recovery, achieving full detection, notification, and restoration within 9.8 μs , with minimal disruption to 5G services. The integration of fast optical switching with precise timing control significantly extends the operational capabilities of the fronthaul. It enables energy-efficient operation and dynamic adaptability, contributing to a more flexible and resilient RAN architecture aligned with future 6G requirements.

2. BACKGROUND AND RELATED WORK

A. Ethernet-Based Solution for Fronthaul Networks

With the disaggregation of RAN functions and the adoption of software-defined and virtualized architectures, Ethernet has emerged as a widely considered transport technology for fronthaul networks. Compared to traditional common public radio interface (CPRI), Ethernet offers cost-effectiveness, ubiquity, and compatibility with commodity hardware, making it an attractive option for scalable Open RAN deployments. To reduce transport bandwidth requirements and enable Ethernet-based fronthaul, the enhanced CPRI (eCPRI) specification was introduced [25]. eCPRI supports packet-based fronthaul by transmitting only active IQ data over Ethernet. IQ data refer to the in-phase and quadrature components of the modulated radio signal, represented as complex samples that carry amplitude and phase information for baseband processing, significantly reducing bandwidth consumption and enabling statistical multiplexing. However, eCPRI still inherits strict latency and jitter constraints—particularly under high PHY-layer split options such as 7.2 $\times$, which typically assume one-way end-to-end transport latency on the order of 100–250 μs between DU and RU, with very tight jitter bounds to preserve HARQ timing and synchronization. This means that even tens of microseconds of additional delay or jitter introduced by intermediate switches can already consume a significant fraction of the available budget. Some implementations have explored the use of Ethernet switches for fronthaul reconfiguration. For example, the fronthaul slicing architecture (FSA) uses dataplane-programmable switches [26]. While promising, such use cases remain limited to coarse link-level

reconfiguration due to Ethernet's inherent queuing delays, variable buffering, and traffic contention—especially under dynamic loads. Conventional store-and-forward Ethernet switches contribute a few microseconds of fixed forwarding latency per hop under light load, but the shared queues can add tens to hundreds of microseconds of variable delay and jitter during contention, which conflicts with the above fronthaul latency and jitter constraints in multi-hop scenarios.

To improve determinism in Ethernet transport, FlexE has been proposed as a potential solution for supporting full packet-based xHaul networks. FlexE introduces converged client interfaces that support a range of Ethernet MAC rates and provide increased flexibility for Ethernet-based service chains. By employing time-division multiplexing (TDM) calendars within the FlexE shim, bandwidth can be finely allocated across large physical Ethernet links, offering enhanced isolation between services and support for network slicing [27]. However, FlexE remains a physical layer management technology, and the maturity of corresponding switching mechanisms is still limited. FlexE operates at the physical layer by creating fixed bandwidth channels over Ethernet links. Unlike packet-based switching, FlexE switching is currently limited to static pipe configuration, with minimal flexibility for scenarios requiring rapid bandwidth reallocation or dynamic traffic engineering. While it addresses bandwidth isolation and service convergence, low-latency switching—critical for RAN fronthaul—is not yet fully supported [13]. In practice, FlexE calendars are configured on timescales of seconds or longer and are not intended for sub-millisecond, event-driven switching of individual fronthaul paths. TSN is another promising approach that seeks to enable deterministic Ethernet transport [28]. Originally developed for industrial and internet applications, TSN has evolved to include enhancements such as traffic shaping, time-aware scheduling, and standardized time synchronization [29]. TSN time-aware scheduling can keep the latency of scheduled high-priority flows in the order of a few tens of microseconds with bounded jitter, but the schedules are pre-computed and updated on millisecond-or-longer timescales, which prevents ns- or microsecond-scale reconfiguration in response to individual failures or DU-migration events. These electrical packet-switching solutions can only reconfigure paths when multiple link options are available. Due to the unpredictable latency and queuing characteristics of Ethernet switches, implementing multi-hop fronthaul networks with these technologies is not yet feasible. Additionally, these electrical switching solutions face challenges in fronthaul networking because of the necessary OEO conversions and the associated management complexity.

In summary, Ethernet technologies such as eCPRI, FlexE, and TSN can deliver fronthaul streams with end-to-end one-way network latency on the order of 100 μ s and microsecond-scale jitter when paths are statically engineered, but their reliance on buffering, pre-configured calendars, and millisecond-scale control loops means they cannot provide the real-time (sub- μ s to ns-scale) reconfigurability and low-latency switching required for dynamic DU migration and fast fronthaul recovery.

B. Optical Switching-Based Solutions for Fronthaul Networks

Optical switching has gained increasing attention as a promising solution to bring sub-channel flexibility, inspired by Google's recent deployment of optical circuit switching (OCS) in their tensor processing units (TPUs) [30]. Unlike traditional electronic switches, optical switches eliminate power-hungry OEO conversions at intermediate nodes, thereby reducing energy consumption, latency, and cost; however, they lack memories for packet-level processing [31]. These properties make optical switching more suitable for circuit-level switching in DCNs, as a complement to electrical packet switching systems [32]. Usually, these applications require optical switches with switching time at the sub-millisecond level. Fast optical switches, particularly those based on semiconductor optical amplifiers (SOAs) [33] and photonic integrated circuits (PICs) [34], have demonstrated ns-scale switching times. Some initial explorations in DCNs demonstrated high throughput, flexible topology reconfiguration, and fine-grained traffic management [21–23].

Beyond DCNs, several works have explored optical switching-based architectures in 5G/6G fronthaul. Xu *et al.* have reviewed advanced radio-over-fiber interfaces and high-speed optical links for large-capacity 6G fronthaul, highlighting wavelength- and sub-channel-granular optical switching as enablers for scalable, low-latency DU–RU interconnection [35]. Yang *et al.* have proposed adaptive time-slot optical switching for latency-sensitive IoT fronthaul, using time-division optical switching to guarantee bounded delay for selected flows [36]. Both approaches demonstrate the potential of optical switching to support flexible fronthaul connectivity but rely on microsecond- to millisecond-scale reconfiguration and scheduled time slots rather than the ns-scale, event-driven path changes targeted in this work. Recent surveys on optical x-haul [37] and 6G optical fronthaul [12] further classify these works as part of a broader move toward reconfigurable optical transport, while also identifying control-plane scalability, synchronization, and protection switching as open challenges.

However, bringing fast optical switching into fronthaul networks introduces new challenges. First, clock management becomes critical: achieving coordinated ns-level switching across multiple nodes requires precise time synchronization. Existing optical switching systems often rely on distributed timing signals delivered over dedicated links [23], but this approach is not scalable or robust to dynamic topology changes and link failures, with much longer coverage in fronthaul than in typical DCN use cases. The distributed clock with a dedicated clock channel experiences different delays compared to the data path, and it lacks the ability to continuously track these variations. Second, the control plane (SDN controller) remains a bottleneck to managing multiple optical switches. Traditional SDN or network orchestration mechanisms are not equipped to handle the rapid control cycles required for ns-scale switching. The absence of fast, coordinated control logic limits the responsiveness and scalability of OCS in large or highly dynamic environments, such as those anticipated in 6G RAN deployments. Third, total reconfiguration time—including both hardware switching latency and control

overhead—is a critical factor for real-world performance. To support scalable and real-time applications, this reconfiguration time must remain independent of network scale, as highlighted in prior studies [23,38,39]. The proposed architecture operates at the transport layer, making it compatible with current 5G and future 6G fronthaul requirements. Integration requires fast optical switches, PTP-capable timing modules, and a time-aware SDN controller, while existing DU/RU configurations remain unchanged. The architecture leverages Ethernet-based PTP synchronization, which can reuse existing network interfaces, and SyncNet can be integrated with standard SDN frameworks through APIs, enabling interoperability with orchestration and management systems. Therefore, practical optical switching solutions for fronthaul must not only offer fast switching but also guarantee synchronized, loss-minimizing operation across heterogeneous and reconfigurable network topologies. These challenges motivate the need for new architectures that combine fast optical hardware with precise, scalable synchronization and a time-aware control plane—capabilities that are largely missing in current optical switching frameworks.

3. PRINCIPLE OF FRONTHAUL NETWORKS BASED ON FAST OPTICAL SWITCHES

The principle of the proposed fronthaul network is illustrated in Fig. 1. Instead of relying on fixed point-to-point connections between RUs and DUs, we introduce a reconfigurable optical fronthaul network enabled by fast optical switches. This dynamic fronthaul architecture allows flexible reconfiguration of interconnections between DUs and RUs, enhancing RAN operations from two key perspectives. In a fronthaul network, any RU can connect to the chosen DU through multiple paths, offering flexibility that surpasses mere reconfigurability.

First, it enables RU traffic migration, whereby multiple RUs can be aggregated onto a consolidated DU during periods of

low traffic (e.g., at night), reducing overall power consumption at the DU level. When traffic increases, the RU can be reconfigured with additional DUs to enhance capacity. Second, the reconfigurable fronthaul introduces link resilience with available multiple paths, allowing for near-instantaneous recovery from link failures, thereby improving the reliability and robustness of wireless network services.

These two scenarios—DU consolidation for energy efficiency and rapid recovery from fronthaul failures—are experimentally validated later in this paper.

However, upgrading from static fronthaul links to a reconfigurable fronthaul network poses significant challenges, primarily due to the stringent latency requirements inherent to fronthaul transport [40]. Conventional Ethernet switches introduce a fixed forwarding latency in the microsecond range per hop, but, more critically, their shared buffers can add tens to hundreds of microseconds of additional, non-deterministic queuing delay under contention. In multi-hop fronthaul topologies, this stochastic queuing behavior can easily consume a 100–250 μ s latency budget and violate the tight jitter requirements, even when TSN or FlexE enhancements are employed. Traditional optical fronthaul deployments based on CPRI over dedicated fiber or WDM already provide essentially deterministic, propagation-limited latency and very low jitter, and therefore meet the timing constraints discussed above. However, these static point-to-point optical links do not offer reconfigurable multi-hop networking or energy-aware consolidation of RUs and DUs, which are the main targets of this work. To address this, we adopt ns-scale fast optical switches that eliminate the need for OEO conversion and enable instant link configuration under synchronized control. This fast optical switching approach is key to achieving both the required latency performance and operational flexibility in future RAN architectures. The key enabling technologies for fronthaul networks will be reported in the following chapter.

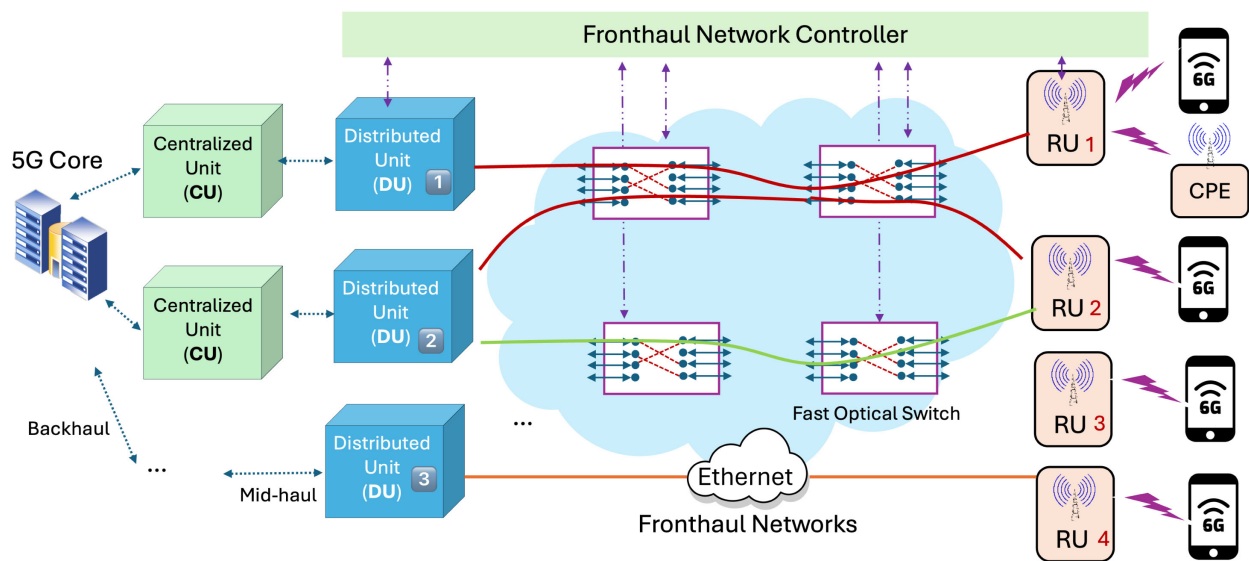


Fig. 1. Principle of fronthaul networks based on fast optical switches in radio access networks. The red paths indicate primary fronthaul routes, while the green paths indicate alternative protection routes, and the orange lines indicate the static links between the distributed units (DUs) and remote units (RUs). CPE, customer premises equipment.

4. DEVELOPMENT AND LAB VALIDATIONS OF KEY ENABLING TECHNOLOGIES FOR FRONTHAUL NETWORKS

To realize the proposed flexible and resilient fronthaul network, we developed and integrated several enabling technologies, including ns-scale optical switches with local control agents (including SDN agent + timing module), a precise time synchronization system for multi-switch coordination, and an SDN controller capable of real-time management and dynamic reconfiguration.

The fast optical switch enables near-instantaneous path changes, essential for link recovery and DU migration. Reconfiguring links in fronthaul networks require coordination of multiple ns optical switches. We implemented a PTP-based synchronization scheme over Ethernet to coordinate switching events across distributed nodes, achieving alignment within sub-100 ns. The SDN controller, named the “SyncNet” controller, incorporates real-time telemetry for rapid failure detection and supports both reactive and scheduled switching with ns precision. We validated these technologies in a controlled lab testbed.

A. Timed Nanosecond Optical Switches

Figure 2 illustrates the overall architecture of the timed ns optical switch developed for fronthaul networks. The core switching device is a 1×4 magneto-optic solid-state optical switch based on a beam-steering mechanism enabled by electrically induced polarization rotation [41]. This device, NanoSpeed Cascaded 1×4 Fiberoptic Switch (model number NPNS-4A21953331) sourced from Agiltron Inc., achieves rapid switching through a compact magneto-optic solution with a fast-response driver requiring three SMA ports to control switching.

To enable precise and programmable control of switching events, we developed a local control agent, named the timed controlling agent in Fig. 2, that interfaces with the central SDN controller. The control agent is built around a single-board computer (e.g., Raspberry Pi) running embedded

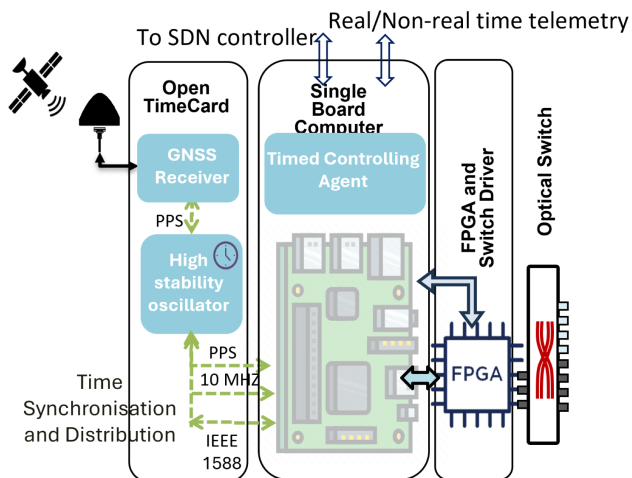


Fig. 2. Design of the timed nano-second optical switch with a local controlling agent.

Linux, which connects to the control plane (via Ethernet and synchronizes its local clock using the PTP).

To achieve ns-level accuracy, the control agent incorporates an Open Timecard module (Timebeat Inc.). This module features a high-stability oscillator and supports synchronization via GNSS/GPS signals or a master clock using PTP. The Open Timecard provides a foundational element for precise local time, synchronized either over a network or directly from a GNSS/GPS source. It integrates a GNSS receiver to deliver highly accurate UTC-aligned pulse-per-second (PPS) and NMEA time signals. NMEA refers to the National Marine Electronics Association standard, which defines protocols for transmitting GPS and time information over serial interfaces. A high-stability oven-controlled crystal oscillator (OCXO) ensures sub-ns phase stability, maintaining robust timing even during GNSS outages and updates. A PCIe interface facilitates high-throughput timestamping and seamless integration, making the Open Timecard a reliable source for deterministic, time-sensitive operations. The module's key features are its 1 PPS output and 10 MHz reference clock, which provide a high-accuracy timestamp for connected components, achieving high precision synchronization accuracy through PTP (IEEE 1588v2). The high-stability oscillator ensures precise timekeeping for the control agent. The Timecard delivers a precise 1 PPS signal to regulate and monitor time differences between multiple nodes, accounting for link changes. It also serves as the timing reference for scheduling switching operations. This controller can be integrated with higher-level network management platforms for unified control. To support monitoring, we developed a Grafana-based dashboard that visualizes synchronization performance and timing offset in real time. Combined with FPGA-based local agents, this approach enables precise, automated control of the optical fronthaul network while maintaining compatibility with existing operational workflows.

Switching events are controlled using an FPGA board, based on the Xilinx Virtex UltraScale platform, which receives the 1 PPS reference and internally generates phase-locked 10 MHz timing signals through a tightly coupled phase-locking loop (PLL) system. This FPGA enables future-time scheduling of switching actions with high temporal precision by generating trigger signals aligned to pre-defined transmission slots.

The result is a fully synchronized switching system across distributed nodes, capable of executing coordinated switching events at ns resolution. Figure 3(a) shows the measured rise time of the optical switch, demonstrating a switching transition time of approximately 10 ns, which validates its suitability for low-latency fronthaul reconfiguration. The synchronized switching will be reported in the following section.

B. PTP-Based Time Synchronization for Coordination of Multiple Optical Switches

To fully exploit ns-level optical switching, switches deployed at different network locations must be precisely coordinated. For example, two separate switches must execute reconfiguration commands at exactly the same scheduled time to ensure seamless connectivity changes, which leverage the benefit of fast switching time. Achieving this level of coordination requires

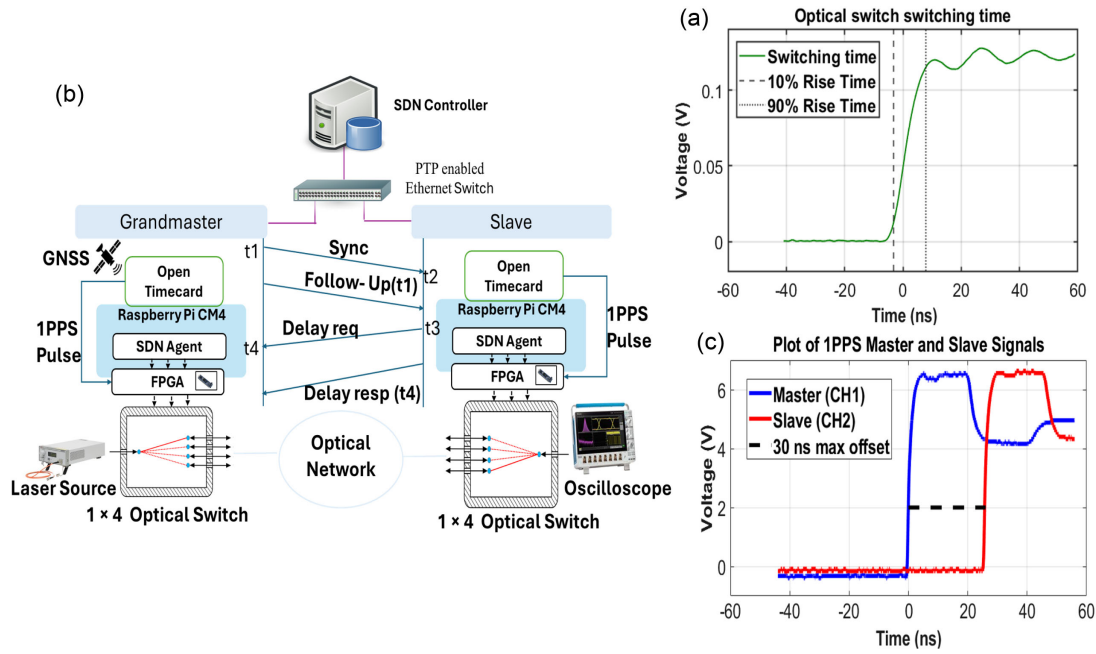


Fig. 3. PTP-based synchronized operations of two optical switches. (a) Measured rising time of the 1×4 optical switch. (b) Optical switches and PTP-based time synchronization. (c) Observed maximum delay between two PPS signals from two Open Timecards.

high-accuracy time synchronization across all participating nodes. In our design, synchronization is achieved using the PTP over Ethernet connections. The control agents described in Section 4.A serve as PTP endpoints, aligning their internal clocks via PTP and maintaining stability through the disciplined oscillators integrated in the Open Timecard modules. In the current implementation, synchronization focuses on aligning absolute time rather than phase and frequency. The local oscillators on the Open Timecards maintain this synchronized time and provide the precise triggers required for ns-level switching events.

The experimental setup, shown in Fig. 3(b), was designed to evaluate synchronization accuracy and jitter under different network conditions. Each node in the system consisted of a Raspberry Pi CM4 running the control agent and acting as the local controller, an Open Timecard for precise time management, and an FPGA responsible for generating high-speed switching commands to the 1×4 optical switch. The designated primary (grandmaster) and secondary (slave) nodes were synchronized through PTP, with absolute time obtained from GPS. A central SDN controller managed the operation of both nodes through its southbound interface, issuing configuration commands to the control agents, which then instructed the FPGAs to execute the scheduled switching events.

To measure the time jitter between two synchronized nodes, we implemented a waveform-based method using optical pulse generation. A continuous-wave (CW) laser was launched sequentially through two optical switches, each located at a different remote node. The combined optical output was detected using a high-speed photodiode (PD) connected to a real-time oscilloscope. In this setup, the first switch controlled the rising edge of the pulse, while the second switch controlled the falling edge, thereby defining the pulse width. The pulse

width variations reflect the jitter of the synchronized system. To enable this measurement, both optical switches were configured with a 200 ns switch-on window, with a deliberate 100 ns temporal offset applied between them. The resulting overlap between the two windows produced an optical pulse of approximately 100 ns duration. This temporal alignment was achieved using two synchronized Open Timecard modules, each operating at one of the remote nodes and disciplined via PTP to a common master clock. The start times of each switch-on event were controlled using 1 PPS (pulse per second) signals generated by the respective timecards, ensuring accurate and repeatable timing alignment. As shown in Fig. 3(c), the maximum observed temporal offset between the master and slave PPS signals was approximately 30 ns. A locally generated 250 MHz clock aligned with the PPS signals allows fine time alignment at a 4 ns resolution. The resulting optical pulses were captured and analyzed to quantify timing jitter. As illustrated in Fig. 4(a), this configuration produced optical pulses with a 100 ns pulse width. To measure the precise jitter measurement, we plotted a statistical histogram of the falling edge, as shown in Fig. 4(b); the jitter is determined by the width parameter of the histogram plot, which is measured to be 30 ns. Notably, this value is consistent with the maximum offset previously measured between the master and slave devices. This value reflects the tracking precision of the PTP-based synchronization system under the experimental conditions and demonstrates its suitability for coordinating ns-scale switching events across geographically distributed nodes.

The Open Timecard also provides their internal PTP monitoring system with key performance metrics visualized through Grafana. Continuous measurement of the grandmaster offset relative to the GPS source, as represented in Fig. 4(c), is critical

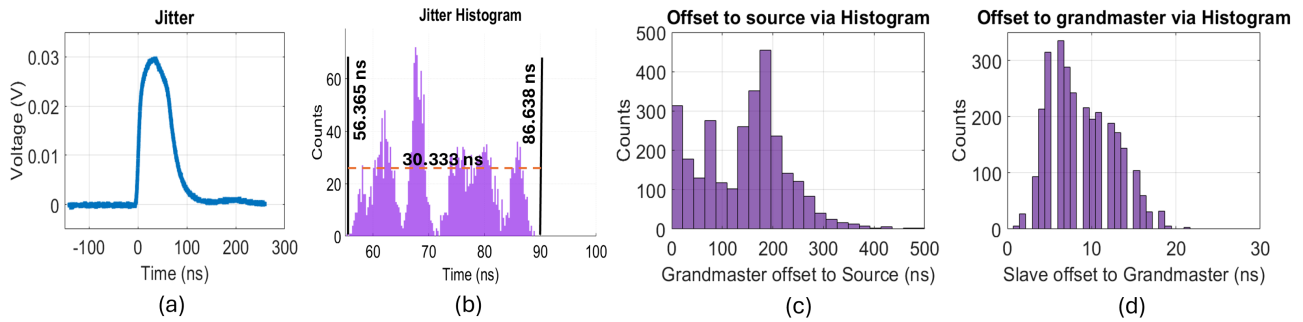


Fig. 4. Jitter measurement. (a) Generated optical pulse with a pulse width around 100 ns for jitter measurement. (b) Distribution of time offsets (ns) for jitter measurement and timing offset distribution. (c) Primary (Grandmaster) offset relative to the source. (d) Secondary (slave) offset relative to the grandmaster.

to ensure the reliability and traceability of the entire time-distribution framework. The GPS system provides an absolute global time reference, ensuring a universal time standard and accuracy of the overall synchronization architecture. The histogram in Fig. 4(d) illustrates the offset of the slave relative to the grandmaster. Here, the distribution is considerably narrower, centered around 7–10 ns, with the majority of samples confined within 10 ns of the mean and 30 ns maximum. This result highlights the improved stability and tighter synchronization performance achieved at the slave, demonstrating that, while the source exhibits fluctuations on the order of hundreds of ns, the final slave alignment to the grandmaster remains within tens of ns.

C. Timed SyncNet Controller

To achieve deterministic ns-level control across the optical network, we integrated PTP-based time synchronization directly into the SDN control plane. A customized controller, referred to as “SyncNet,” was developed to enable time-aware coordination of distributed network devices. “SyncNet” determines the time offset between itself and each control agent using PTP, allowing it to issue switching commands that are interpreted in the local time context of each agent. This design removes the need for direct peer-to-peer synchronization between agents while preserving precise alignment of control actions.

On each node, the control agents use this timing reference to trigger reconfiguration events on their associated optical switches via FPGA interfaces. This architecture supports both immediate, reactive switching and scheduled, time-triggered events. By embedding precise timing into the control framework, “SyncNet” elevates time to a native control dimension in SDN operations. This enables deterministic, low-latency responses for scenarios such as fast failure recovery, traffic-aware reconfiguration, and coordinated optical switching at the ns scale—capabilities that are essential for resilient and flexible fronthaul networks.

In a larger deployment, the latency and jitter between the SDN controller and the distributed agents become a critical factor for overall system performance. While the proposed architecture relies on local time synchronization at each node and scheduled actions referenced to the local clock, excessive control-plane latency can still impact failure detection,

configuration dissemination, and the time needed to install new schedules. If controller–agent round-trip times increase from sub-millisecond values, as in our lab testbed, to several milliseconds or more in a wide-area deployment, the total protection or reconfiguration time would grow accordingly, even though the underlying optical switching still operates at the ns scale. Therefore, practical deployments should either keep the controller logically and physically close to the fronthaul domain or employ hierarchical and distributed control so that most time-critical decisions are executed by local agents, while the central controller focuses on slower re-optimization tasks.

5. FULL SYSTEM VALIDATION OF FRONTHAUL NETWORKS

The experimental setup for validating the proposed network is illustrated in Fig. 5. The Open RAN testbed is based on parallel wireless’s OpenRAN platform, which integrates CU, DU, and RU functionalities along with their respective management modules on dedicated servers.

All radio access components, including the RU and the 5G antenna, are supplied by parallel wireless, supporting licensed-spectrum, standalone (SA) 5G connectivity to end-user devices such as smartphones and customer premise equipment (CPE).

To support a reconfigurable fronthaul segment, the traditional direct fiber connection between the DU and RU has been replaced with a dual-path optical transport architecture. This setup includes two 1×4 optical switches dedicated to uplink and downlink directions for dynamic real-time path selection. Each direction features parallel fiber links: a primary path of 25 km and a recovery path of 2 km, both equipped with 10 Gbps SFP optical transceivers capable of reaching up to 40 km for high-capacity transmission. On the downlink, a PD is interfaced with an FPGA to enable real-time optical signal loss detection. Upon detecting signal loss, the FPGA promptly reports this event to the SDN controller, which orchestrates fronthaul reconfiguration.

The fronthaul control and management plane is synchronized through the PTP-based timing framework described in Section 3, ensuring alignment among all switching nodes. At the heart of this control solution is a time-aware SDN controller, which orchestrates the FPGAs managing the optical switches. In response to link degradation, power loss, or traffic

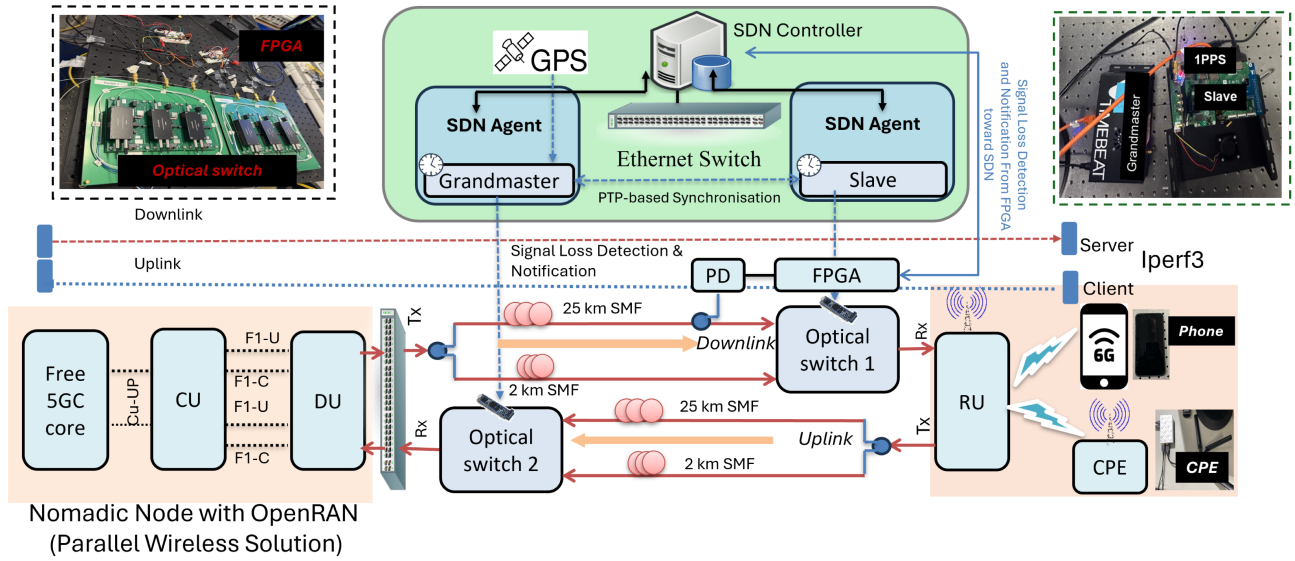


Fig. 5. Experimental setup of the Open RAN system with proposed fronthaul networks for validation.

dynamics, the controller can autonomously trigger the switches to implement either a recovery action (rerouting traffic to the backup link after a failure) or a planned reconfiguration (migrating fronthaul paths based on traffic/load), both without human intervention. This dual capability ensures the fronthaul is both resilient to failures and adaptable to changing operational requirements.

A. Network Reconfiguration

The reconfiguration scenario demonstrates the capability of the proposed fronthaul network to support dynamic migration of RUs when they are underutilized or idle, thereby improving overall energy efficiency in RAN. This is achieved by dynamically switching both uplink and downlink traffic to alternate optical paths, effectively consolidating active RUs onto fewer DUs during periods of low traffic demand. The simultaneous reconfigurations minimized the disturbance to 5G services.

In this scenario, the system performs reconfiguration by rerouting fronthaul traffic through a predefined recovery path using 1×4 optical switches, while the original primary path is temporarily freed. The reconfiguration process is carefully orchestrated to ensure session continuity and seamless connectivity, with no packet loss observed during switching events.

Experimental validation in Fig. 6(a) shows that, following the manual initiation of RU migration, throughput temporarily decreases as the system transitions paths. However, the network rapidly stabilizes, restoring up/downlink transmission rates to near-baseline levels with minimal fluctuation. Importantly, the RU–CPE connection remains active throughout this process without requiring session reinitialization, confirming the effectiveness of coordinated path switching in maintaining uninterrupted service.

The oscilloscope traces in Figs. 6(c) and 6(d) presented the well-defined transmission window and the ns-scale

transition time (10 ns rise time) during link reconfiguration, demonstrating the scheme's suitability for high-speed reconfiguration.

This dynamic migration approach underscores the flexibility of the optical fronthaul architecture to perform energy-aware resource reallocation, allowing the network to adapt dynamically to traffic patterns by consolidating RUs and reducing power consumption without compromising reliability or quality of service.

B. Fronthaul Network Recovery

The network recovery scenario evaluates the fronthaul's ability to restore optimal performance after a link failure, focusing on achieving seamless service continuity with minimal data loss during automatic rerouting, as represented in Fig. 6(b).

In this use case, downlink recovery is investigated. To verify the synchronized operations, the coupler in the downlink is replaced with the other optical switch. Upon detecting a link failure via the PD on the downlink, the FPGA promptly signals the SDN controller to initiate an autonomous recovery procedure. The recovery process is carefully timed using the PTP-based synchronization framework described in Section 3. This coordinated switching prevents transient mismatches in traffic routing that could otherwise cause packet drops or ordering issues. This procedure involves synchronized reconfiguration of both optical switches to redirect traffic from the failed primary path to the protected recovery route. In Fig. 6(e), the trace for link recovery indicates a total recovery time of 9.8 μ s. This duration encompasses the link power loss, notification to the SDN controller, and subsequent immediate reconfiguration.

Traffic throughput and packet delivery were continuously monitored during recovery using an iperf3 client-server setup and the MD1230B data quality analyzer, with data traces precisely time-stamped using the synchronized timing system for accurate correlation of network events.

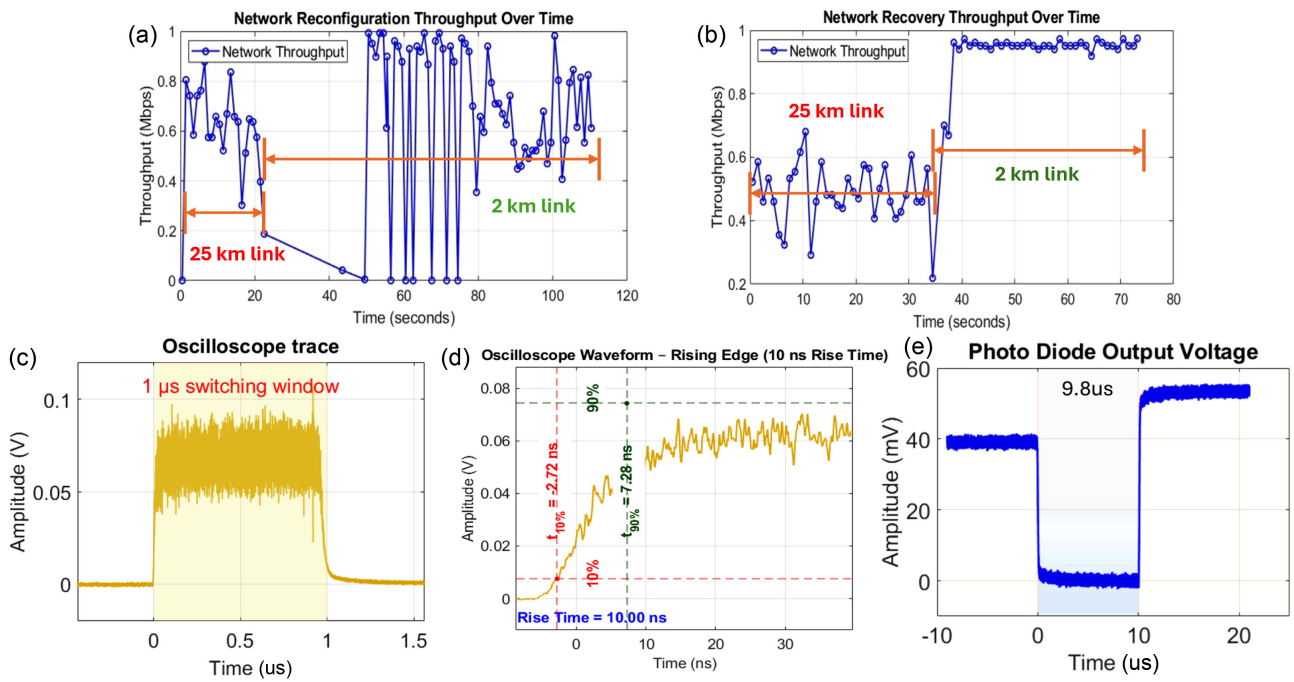


Fig. 6. Network throughput over (a) reconfiguration and (b) recovery; oscilloscope trace (c) switching window, (d) switching window rising time, and (e) link recovery downtime.

Over the recovery interval spanning approximately 80 s, 2,043,059 IPv4 packets were transmitted. Of these, exactly five packets were lost during the switching window, representing an exceptionally low packet loss rate of approximately 0.00024%. A detailed examination of packet capture and timing logs indicates that these lost packets correspond to those in transit at the precise moment of optical path transition—when the switches simultaneously transitioned links, creating a roughly 10 μ s interruption during which the optical signal briefly dropped before stabilization, as represented in Fig. 6(e). This minimal packet loss is validated by correlating the time stamps from the synchronized control plane with packet sequence numbers captured by the analyzer, confirming that all subsequent packets were delivered successfully once the recovery path was established. The tight synchronization between uplink and downlink switching enabled by the PTP-based control framework is critical in confining loss to this brief switching window, effectively preventing prolonged service disruption.

The measured recovery time in our testbed reflects both the ultra-fast optical switching and the relatively low latency between the controller and the agents in a local deployment. In wide-area networks with higher controller-agent latency, the same optical switching hardware and synchronization framework would still provide ns-scale path reconfiguration once commands arrive, but the overall fault-to-recovery interval would be dominated by the propagation and processing delays in the control plane. This highlights the importance of control-plane placement and possibly hierarchical control architectures when scaling the proposed fronthaul solution to larger geographical areas. Building on this work, we have recently integrated an AI-driven control model into the optical switching framework to further minimize effective downtime.

This model predicts agent-switch communication delays and proactively adjusts control actions and scheduling, thereby mitigating the dominant control-plane component of the recovery time and bringing the overall protection behavior closer to the intrinsic ns-scale capabilities of the optical switches.

These results demonstrate the robustness and near-seamless protection switching capabilities of the proposed ns optical fronthaul network, validating its practical suitability for latency-sensitive 5G and future 6G applications requiring ultra-reliable, low-loss recovery.

C. Discussion on Scalability

Fronthaul networks operate under stricter constraints on latency, synchronization, and packet loss tolerance. Even minor packet loss can disrupt PHY-layer functional splits and impact end-user QoS. In this paper, the experimental validation of fast optical switching-based fronthaul networks demonstrates feasibility for small-scale deployments. However, several factors could limit scalability in larger networks, including the coordination of numerous optical switches based on precise timing. Especially, long-distance fiber links and larger node counts will introduce higher jitter, reducing synchronization accuracy. Expanding network size also increases the complexity of the SDN controller, leading to longer reconfiguration times and slower responsiveness, which will potentially affect the network recovery time. These limitations can be mitigated by implementing robust clock synchronization across nodes to minimize jitter and by adopting distributed SDN controller architectures to improve efficiency in large-scale deployment.

6. CONCLUSION

This paper presents a synchronization-enabled optical switching framework for reconfigurable fronthaul networks, with a focus on improving network resilience and enabling energy-aware operation. By integrating PTP-based time synchronization into the control and data planes, we achieved coordinated ns-level optical switching across distributed nodes, with measured timing jitter within approximately 100 ns under hardware-assisted synchronization.

The proposed SDN control architecture supports both scheduled and on-demand reconfiguration, allowing deterministic switching events to be orchestrated with temporal precision. System-level validation using a live Open RAN testbed demonstrated the feasibility of fast fronthaul link recovery, with minimal disruption to ongoing services. In a representative recovery scenario, packet loss was limited to 0.00024% across more than two million transmitted packets, confirming reliable protection switching under fault conditions. Throughout dynamic path transitions, the system sustained consistent throughput performance—averaging 1.03 (uplink) and 1.01 Mbps (downlink)—while maintaining service continuity without session re-initialization. These results indicate the potential of time-synchronized optical switching to support emerging fronthaul use cases where reconfigurability and reliability are critical. While further scalability and integration studies are needed, this work demonstrates a practical step toward time-aware optical transport in Open RAN systems and lays a foundation for future 6G infrastructures requiring low-latency, adaptive networking capabilities.

Funding. Department for Science, Innovation and Technology (REASON); Engineering and Physical Sciences Research Council (ECO-eNET 10113933, HASC EP/X040569/1, CSA Catapult).

Acknowledgment. Partial support was also provided by the CSA Catapult-UoB collaboration project. We extend our gratitude to Ian Gough and Lasse Johnsen from Timebeat for their contributions.

REFERENCES

- J. Guo, M. Wang, H. Yin, *et al.*, "Large language models and artificial intelligence generated content technologies meet communication networks," *IEEE Internet Things J.* **12**, 1529–1553 (2024).
- M. Johnsen, *Developing AI Applications with Large Language Models* (Maria Johnsen, 2025).
- M. Z. Aloudat, A. Aboumadi, A. Soliman, *et al.*, "Metaverse unbound: a survey on synergistic integration between semantic communication, 6G, and edge learning," *IEEE Access* **13**, 58302–58350 (2025).
- ITU, "IMT traffic estimates for the years 2020 to 2030," Report ITU-R M.2370-0 (2015).
- M. El-Moghazi and J. Whalley, "The ITU IMT-2020 standardization: lessons from 5G and future perspectives for 6G," *J. Inf. Policy* **12**, 281–320 (2022).
- W. Olabiyi, "Latency-aware data delivery modes for high-performance SOC-NOC routers," *SSRN* (2024).
- L. M. Larsen, H. L. Christiansen, S. Ruepp, *et al.*, "The evolution of mobile network operations: a comprehensive analysis of open RAN adoption," *Comput. Netw.* **243**, 110292 (2024).
- D. H. Hailu, B. G. Gebrehaweria, S. H. Kebede, *et al.*, "Mobile fronthaul transport options in C-RAN and emerging research directions: a comprehensive study," *Opt. Switching Netw.* **30**, 40–52 (2018).
- O. E. Martínez-Durive, J. Suárez-Varela, J. Omaña Iglesias, *et al.*, "An evaluation of ran sustainability strategies in production networks," in *IEEE International Conference on Computer Communications* (2024).
- S. Shen, J. Han, K. Bardhi, *et al.*, "Unified monitoring and telemetry platform supporting network intelligence in optical networks," *J. Opt. Commun. Netw.* **17**, 139–151 (2025).
- C. Leonelli, "Dynamic resource allocation and energy optimization in 5G open radio access network (O-RAN)," Master's thesis (University of Bologna, 2024).
- A. Fayad, T. Cinkler, and J. Rak, "Toward 6G optical fronthaul: a survey on enabling technologies and research perspectives," *IEEE Commun. Surv. Tutorials* **27**, 629–666 (2024).
- M. Zhang, "Flex Ethernet technology and application in 5G mobile transport network," *China Commun.* **18**, 250–258 (2021).
- D. Koulougli, K. K. Nguyen, and M. Cheriet, "Cost optimization of FlexEthernet over elastic optical network fronthaul design," *IEEE Trans. Netw. Serv. Manage.* **22**, 3120–3137 (2025).
- K. Zambouri, M. Noor-A-Rahim, J. John, *et al.*, "A comprehensive survey of wireless time-sensitive networking (TSN): architecture, technologies, applications, and open issues," *IEEE Commun. Surv. Tutorials* **27**, 2129–2155 (2024).
- P. A. Baziana, "Optical data center networking: a comprehensive review on traffic, switching, bandwidth allocation, and challenges," *IEEE Access* **12**, 186413 (2024).
- V. N. Yokar, A. Mehrpooya, Y. Teng, *et al.*, "Fast link recovery via PTP-synchronized nanosecond optical switching," in *International Conference on Optical Network Design and Modeling (ONDM)* (2025).
- W. Zhao, S. Shen, V. Yokar, *et al.*, "Fibre vibration sensing and localization with cross-correlation based neural network in a practical bidirectional optical transmission system," in *Asia Communications and Photonics Conference* (IEEE, 2025).
- Y. Teng, A. Mehrpooya, R. Yang, *et al.*, "Fast optical switch enabled filterless SDM networks with adaptive topology," in *51st European Conference on Optical Communication (ECOC)* (IEEE, 2025).
- B. Sun, T. Li, V. Yokar, *et al.*, "Nanosecond electro-optic switching with time synchronisation for fronthaul TSN applications," in *51st European Conference on Optical Communication (ECOC)* (IEEE, 2025).
- D. J. Bishop, C. R. Giles, and S. R. Das, "The rise of optical switching," *Sci. Am.* **284**, 88–94 (2001).
- Z. Chai, X. Hu, F. Wang, *et al.*, "Ultrafast all-optical switching," *Adv. Opt. Mater.* **5**, 1600665 (2017).
- X. Xue and N. Calabretta, "Nanosecond optical switching and control system for data center networks," *Nat. Commun.* **13**, 2257 (2022).
- M. L. Nielsen, "Experimental and theoretical investigation of semiconductor optical amplifier (SOA) based all-optical switches," Ph.D. thesis (Technical University of Denmark, 2004).
- CPRI Consortium, "Common public radio interface: ECPRI specification V2.0," O-RAN.SFG.O-RAN-Security-Protocols-Specifications-v02.00 Technical Specification (2019).
- N. Budhdev, R. Joshi, P. G. Kannan, *et al.*, "FSA: fronthaul slicing architecture for 5G using dataplane programmable switches," in *Proceedings of the 27th Annual International Conference on Mobile Computing and Networking* (ACM, 2021), pp. 723–735.
- A. Eira, A. Pereira, J. Pires, *et al.*, "On the efficiency of flexible Ethernet client architectures in optical transport networks [Invited]," *J. Opt. Commun. Netw.* **10**, A133–A143 (2018).
- S. Bhattacharjee, R. Schmidt, K. Katsalis, *et al.*, "Time-sensitive networking for 5G fronthaul networks," in *IEEE International Conference on Communications (ICC)* (IEEE, 2020).
- S. Waldhauser, B. Jaeger, and M. Helm, "Time synchronization in time-sensitive networking," in *Network Architectures and Services* (2020).
- N. P. Jouppi, G. Kurian, S. Li, *et al.*, "TPU v4: an optically reconfigurable supercomputer for machine learning with hardware support for embeddings," *arXiv* (2023).
- G. Patronas, N. Terzenidis, P. Kashinkunti, *et al.*, "Optical switching for data centers and advanced computing systems [Invited]," *J. Opt. Commun. Netw.* **17**, A87–A95 (2025).

32. S. Yan, E. Hugues-Salas, V. Rancano, *et al.*, "Archon: a function programmable optical interconnect architecture for transparent intra and inter data center SDM/TDM/WDM networking," *J. Lightwave Technol.* **33**, 1586–1595 (2015).
33. D. W. Feyisa, B. Shi, R. Kraemer, *et al.*, "Compact 8×8 SOA-based optical WDM space switch in generic InP technology," *J. Lightwave Technol.* **40**, 6331–6338 (2022).
34. Z. Zhang, M. Chen, R. Ma, *et al.*, "Dilated space-and-wavelength selective crosspoint optical switch," *arXiv* (2025).
35. Y. Xu, L. Fei, Y. Zhu, *et al.*, "Advanced radio-over-fiber interfaces and high-speed optical links toward 6G large-capacity fronthaul networks," *J. Opt. Commun. Netw.* **17**, E17–E36 (2025).
36. C. Yang, X. Xue, B. Pan, *et al.*, "Adaptive time-slot optical switching for latency-sensitive IOT fronthaul in 6G networks," *J. Lightwave Technol.* **43**, 10930–10938 (2025).
37. M. Balanici, S. M. Kouhni, M. Aghaeipour, *et al.*, "A roadmap on optical X-haul technologies for next-generation mobile networks," *IEEE Commun. Mag.* **63**(9), 102–108 (2025).
38. D. Chitimalla, K. Kondepudi, L. Valcarenghi, *et al.*, "5G fronthaul-latency and jitter studies of CPRI over ethernet," *J. Opt. Commun. Netw.* **9**, 172–182 (2017).
39. Y.-H. Wen, J.-W. Ho, and K.-M. Feng, "Simultaneous all-optical transparent phase multiplexing/de-multiplexing based on FWM in a HNLF," in *Optical Fiber Communication Conference* (Optica Publishing Group, 2022), paper W4D.1.
40. G. O. Pérez, J. A. Hernández, and D. Larrabeiti, "Fronthaul network modeling and dimensioning meeting ultra-low latency requirements for 5G," *J. Opt. Commun. Netw.* **10**, 573–581 (2018).
41. J. Zhao, "Next generation of passive and reconfigurable fiber optic components," in *IEEE Avionics, Fiber-optics and Photonics Technology Conference* (2007), pp. 62–63.